Secure Email Agent

Computer and systems Department

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IDEA

Background

- Phishing emails are cyberattacks that use deception and social engineering to steal sensitive information
- Attackers aim to collect passwords, financial details, or personal data
- Messages often create urgency, fear, or curiosity to pressure quick action
- Common tactics include fake links, malicious attachments, and spoofed email addresses
- Phishing exploits human behavior, making it a leading cause of data breaches

The Problem

- Humans are the weakest link in organizational security — attackers exploit human trust, curiosity, and urgency rather than system flaws.
- Phishing remains one of the most critical cybersecurity threats worldwide, responsible for the majority of successful breaches.
- Over 90% of cyberattacks (credential theft, ransomware, fraud) start with a malicious email.
- It represents the Delivery stage of the Cyber Kill Chain, where attackers deliver malicious content through emails, attachments, or links.

Project Importance

- Email remains the universal standard for business and personal communication worldwide with over 4 billion users globally, it provides an infinitely scalable attack surface for cybercriminals. Protecting this channel is protecting the backbone of global communication.

- By detecting and blocking the phishing email, we effectively sever the attacker's entry point, preventing them from establishing a foothold to launch malware, ransomware, or steal credentials.

Our solution

The Hook

AI-Powered Email Assistant for Phishing Detection
& Smart Email Management

- Traditional email platforms struggle with phishing prevention, spam accuracy, and inbox prioritization.
- The Hook addresses these challenges by providing users with an AI assistant that can analyze emails, detect suspicious activity, summarize content, and help manage inboxes efficiently.

Survey

Users

High Freq

- Too many spam emails
- “Summarization would save time.”

- Opens links without verification
- “Important emails sometimes go to spam.”

Low Priority

High Priority

- “Nothing really annoys me much.”

- “I always verify the domain before clicking anything.”

Low Freq

Empathy Map

For Email Agent

Says

I just trust the logo.

There are too many spam emails.

I wish the email could summarize messages.

Thinks

Unsure if some emails are safe or malicious

Wants an AI assistant to help manage emails

Does

Check inbox multiple times per day

Sometimes open links without deep verification

Trust the sender without vefications

Feels

Worried about phishing and scams

Vulnerable

User Persona

Core Profile

Name: Ahmed Hassan

Age and gender: 24

Job: Accountent

Email Apps Used: Gmail

Background

Ahmed works in a company where email is the primary communication channel with clients, banks, service providers, and internal teams.

Pain Points

- Important emails sometimes go to spam
- Some email filters are not accurate
- Concern about email security and personal information

Goals

- Identify important emails quickly
- Avoid phishing and malicious emails
- Use a system that automatically filters and organizes emails
- Have an AI assistant to analyze emails and help respond

Needs

- AI agent that filters and replies to emails
- Strong protection against phishing attacks
- Better email classification

Behaviour

Usually checks sender address carefully but sometimes just trusts logos or domains

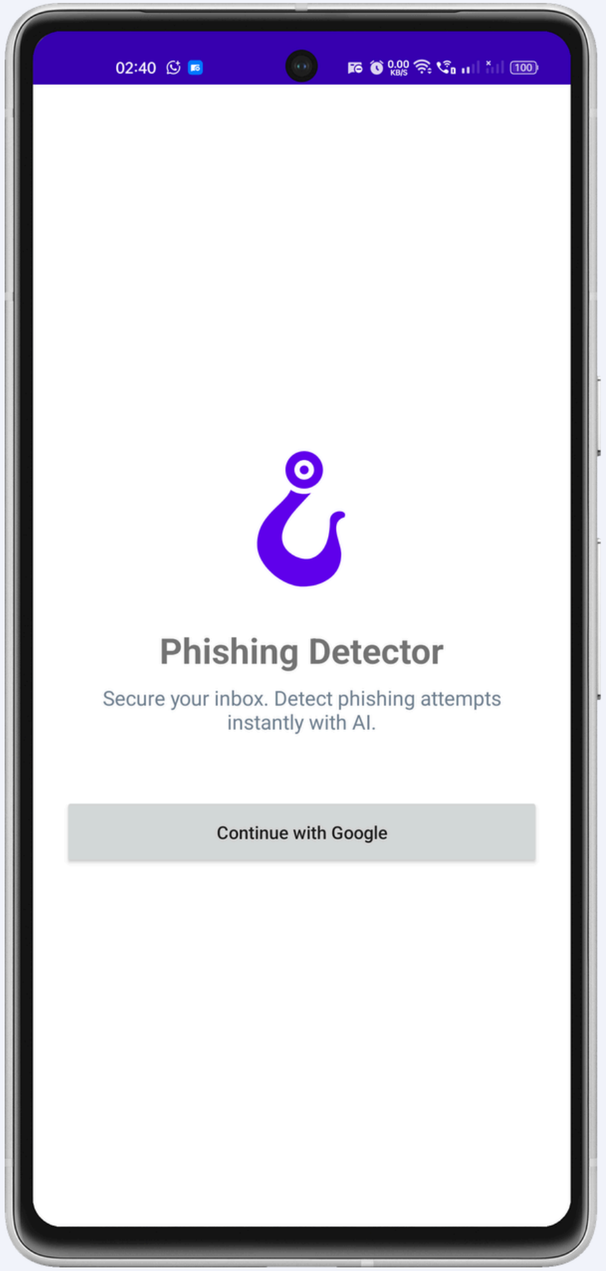
Pain Points & Problem Statement

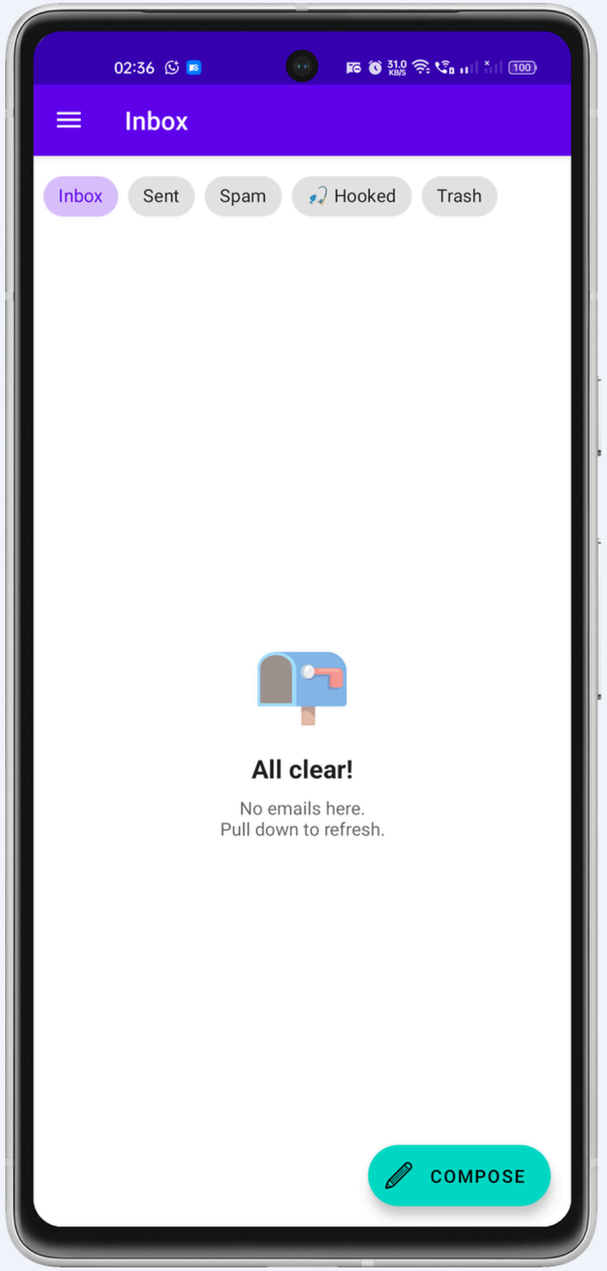
Key Pain Points:

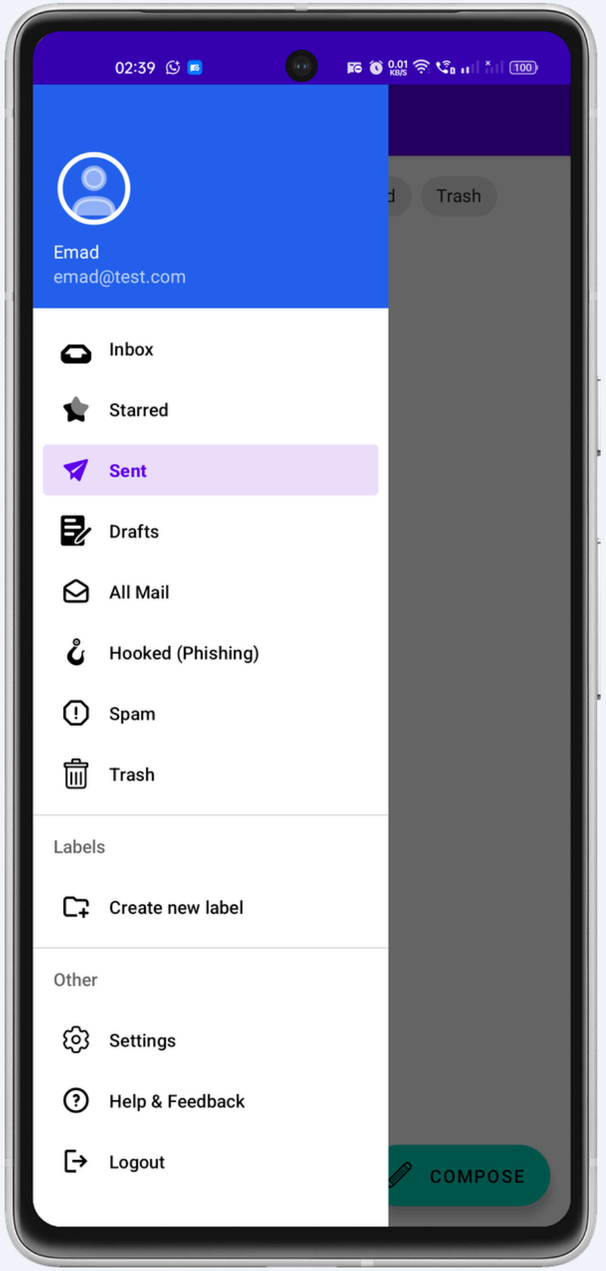
- Inaccurate spam filtering routes important emails incorrectly
- Users vulnerable to phishing through visual branding trust
- Email overload with poor prioritization
- No AI assistance for summarization or smart replies
- Anxiety around malicious links and attachments

Problem Statement: "Busy email users struggle to identify phishing attempts, prioritize important messages, and efficiently manage overloaded inboxes because current email platforms lack intelligent, personalized, and security-focused assistance."

UI


02:40
Phishing Detector
Secure your inbox. Detect phishing attempts instantly with AI.
Continue with Google


02:36
Inbox
Inbox
Sent
Spam
Hooked
Trash
All clear!
No emails here.
Pull down to refresh.
COMPOSE


02:39
Trash
Emad
emad@test.com
Inbox
Starred
Sent
Drafts
All Mail
Hooked (Phishing)
Spam
Trash
Labels
Create new label
Other
Settings
Help & Feedback
Logout
COMPOSE

Email Detail Screen

AM **Ahmed Majed** 10:32 AM
to: me

Project Update: Q3 Deliverables and Roadmap Review

Hi team,

I've attached the final version of the project roadmap for Q3. Please take a look at the key milestones and let me know if there are any conflicts with your current schedules.

We've incorporated the feedback from last Friday's session regarding the user onboarding flow. The priority list remains the same, but we've shifted some internal deadlines to accommodate the design polish phase.

Best regards,
Jane

q3_roadmap_final.pdf
1.2 MB

Reply | Reply All | Forward

Compose Screen

New Message | Send

From: john.doe@gmail.com

To: Recipients

Subject: Subject

Compose email

Hooked Screen (Empty State)

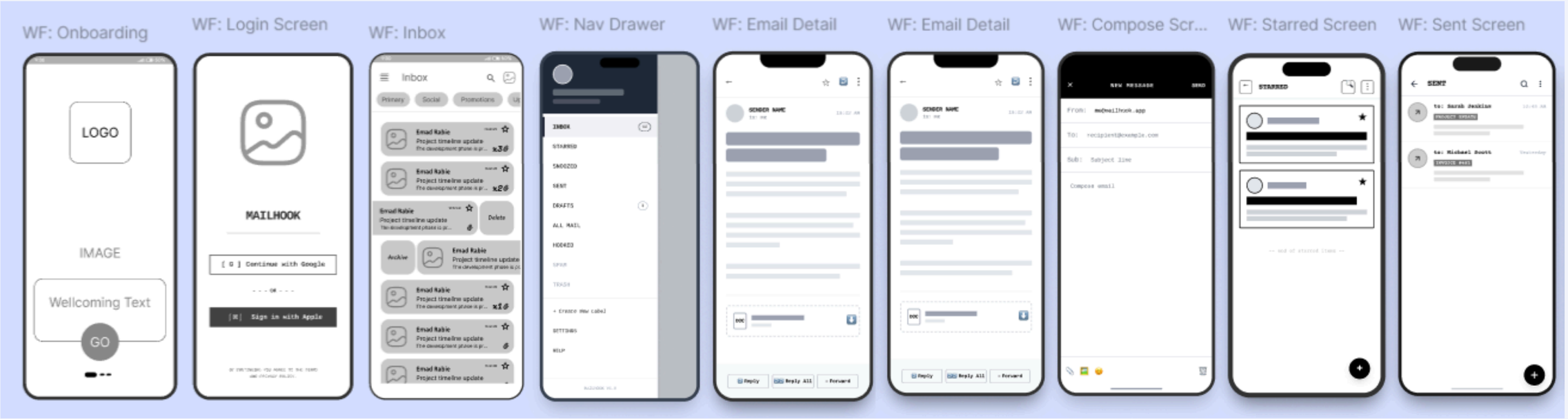
Mailhook | JD

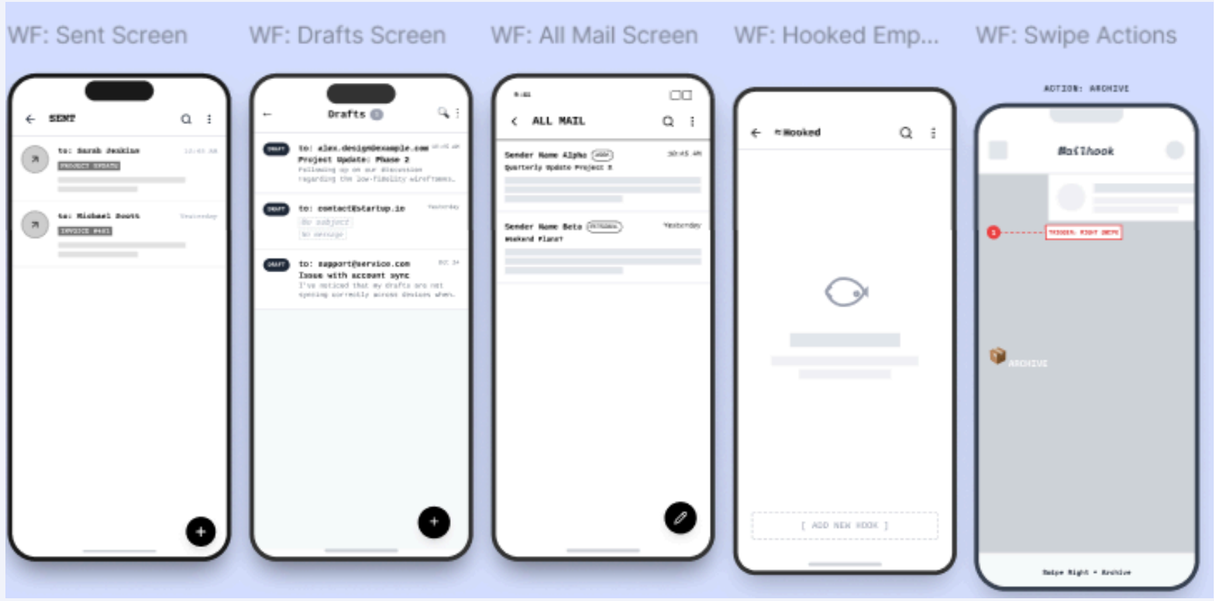
No hooked emails yet.

Tap the hook icon on any email to save important messages here for quick access.

Hook an email

INBOX | HOOKED | SENT | SETTINGS


WF: Onboarding
LOGO
IMAGE
Wellcoming Text
GO
WF: Login Screen
MAILHOOK
[G] Continue with Google
- - - OR - - -
[ ] Sign in with Apple
WF: Inbox
Inbox
Primary
Social
Promotions
Delete
Archive
WF: Nav Drawer
INBOX
STARRED
SNOOZED
SENT
DRAFTS
ALL MAIL
HOOKED
SPAM
TRASH
+ Create New Label
SETTINGS
HELP
WF: Email Detail
SENDER NAME
Reply
Reply All
Forward
WF: Email Detail
SENDER NAME
Reply
Reply All
Forward
WF: Compose Scr...
NEW MESSAGE
SEND
From:
To:
Sub:
Subject line
Compose email
WF: Starred Screen
STARRED
WF: Sent Screen
SENT


WF: Sent Screen
SENT
WF: Drafts Screen
Drafts
WF: All Mail Screen
ALL MAIL
WF: Hooked Emp...
Hooked
[ADD NEW HOOK]
WF: Swipe Actions
ACTION: ARCHIVE
Mailhook
ARCHIVE
Swipe Right = Archive

Primary

Primary	Primary Light	Primary Dark
#2563EB	#DBEAFE	#1D4ED8

Neutrals

Text Primary	Text Secondary	Text Muted	Placeholder	Divider	Background	Card BG
#111827	#374151	#6B7280	#9CA3AF	#E5E7EB	#F8FAFC	#FFFFFF

Accents

Star / Hook	Delete / Draft	Archive	Spam / Warn
#F59E0B	#EF4444	#10B981	#F97316

Avatar Colors

JD #7C3AED	MS #059669	DM #D97706	AL #DB2777	JD #2563EB
RK #DC2626				